EE2025: Engineering Electromagnetics

Tutorial 5: Waveguides

July-Nov 2025

- Use appropriate approximations and work with reasonable assumptions wherever necessary.
- Good familiarity with Maxwell's equations helps.

Waveguides

1. In this course, we have seen the phenomena of dispersion occur in two separate occasions, first, with rain drops, and second, with waveguides. Are the *physical* reasons for dispersion the same in both cases? Elaborate. What parameters control the amount of dispersion in both cases?

   In a parallel plate waveguide, which modes shows dispersion, and which one is non-dispersive and why?

2. A parallel-plates waveguide has a height of 2 cm and is filled with a dielectric of relative permittivity 2.25. The wave propagates in the z-direction as shown below. The magnetic field inside the waveguide is characterized by

   $$H_z = 50\cos(100\pi x)\cos\left(3\pi \times 10^{10}t - \beta z\right)\ A/m$$

   Determine:

   1. The mode that is propagating
   2. The Phase constant

3. Consider a rectangular waveguide with dimensions 2.28 cm × 1.01 cm. What modes will propagate in this wave guide, if the driving frequency is 17 GHz ? Suppose you wanted to excite only one TE mode; what range of frequencies could we use? What are the corresponding wavelengths (in free space)?

4. An air-filled rectangular waveguide operates at 1 GHz and has $a = 5cm$ and $b = 2cm$. Show that the TM_{21} mode cannot propagate at this frequency.

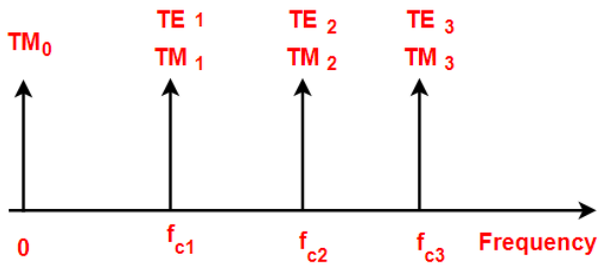
5. Calculate values of β, v_p, λ_g for a 2.5 cm by 1.5 cm rectangular waveguide operating at 7.5 GHz, if the waveguide is air-filled.

6. A section of a rectangular waveguide of cross-section 2 cm × 1.5 cm is to be used as a delay line in a radar at 10 GHz. What should be the length of the section to realize a delay of 10 nsec ?

7. An air-filled section of an X-band (8.2–12.4 GHz) rectangular waveguide of length l is used as a delay line. The longer dimension of the waveguide is a cm and it operates at TE_{20} mode. Given that, $l = 50$ cm provides a delay of 2 ns at 10 GHz, find a.

8. A parallel plate waveguide has a separation $d = 1$ cm, and is filled with teflon, having dielectric constant $\epsilon_r = 2.1$. Determine the maximum operating frequency such that only TEM mode will propagate. Also find the range of frequencies over which TE_1 and TM_1 (m=1) modes will propagate, and no higher order modes.

9. In the previous question, if the operating frequency is 150 GHz, how many waveguide modes will propagate?



Cut-off frequencies for different TE and TM modes

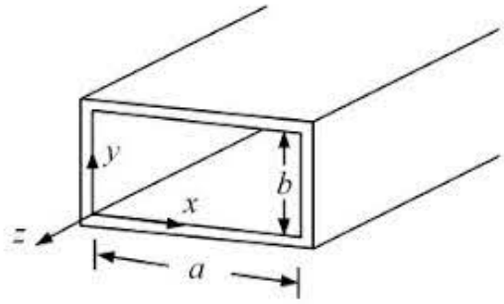
10. In the waveguide mentioned in the previous question, if the operating frequency is 25 GHz, then the modes m = 1 and m = 2 are the only allowed modes (apart from TEM). Determine the group delay difference (the difference in arrival times between different modes) between these two modes over a distance of 1 cm.

    Technically, even the TEM mode is a propagating mode. Find out which pair of modes has the largest group delay over the same distance.

11. For the fundamental mode inside a rectangular waveguide the logitudinal magnetic field is given as

$$H_z = 20\cos(10y)e^{-j\beta z} A/m$$

    Find the cut-off frequency of the mode. Also, find the frequency at which the group velocity is 1/3 of the phase velocity.



12. For a perpendicularly-polarized wave getting reflected from a PEC (perfect electric conductor) at $x = 0$, if another PEC is placed at a distance $x = d$ such that that tangential electric field goes to zero, wave propagates and it is a parallel plane waveguide. Suppose, instead of a PEC, what if we place a PMC (perfect magnetic conductor) at $x = d$. Obtain the value(s) of d for which the wave propagates. (Hint: Analogous to PEC, the tangential magnetic fields are zero on PMC).

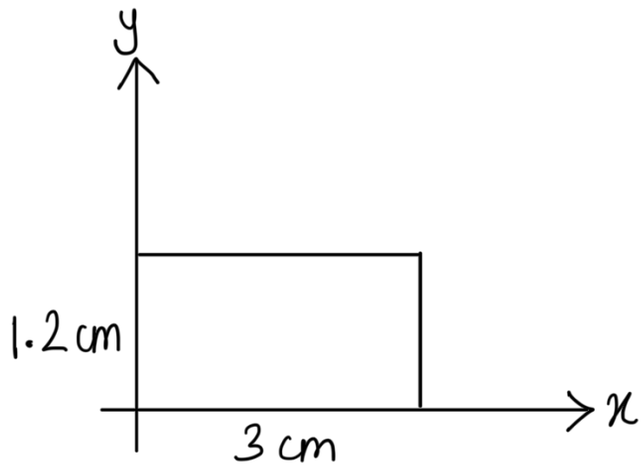
13. A broad class of non-magnetic dispersive media exhibit relative permittivity, ϵ_r, that varies as the square of frequency over a narrow range of frequencies centered at ω_0. For these media we presume

$$\epsilon_r = K\left(\frac{\omega}{\omega_0}\right)^2,$$

    where K is a real valued positive constant. What is the phase and group velocity for a plane wave in this material?

14. The magnetic field along the propagation direction inside a air-filled rectangular waveguide with the cross section shown in the figure is given by:

$$H_z = 3\cos\left(2.094 \times 10^2 x\right)\cos\left(2.618 \times 10^2 y\right)\cos\left(6.283 \times 10^{10} t - \beta z\right)$$



Reason out whether the H field given is feasible for this waveguide.

15. A rectangular waveguide made of PEC has the following characteristics:

$$b = 1.5\,\text{cm},\ a = 3.0\,\text{cm},\ \mu_r = 1,\ \varepsilon_r = 2.25$$

(a) Calculate the cutoff wavelength and frequency for the TE_{10}, TE_{20}, and TM_{11} modes.

(b) Calculate λ_g (guided wavelength) and (Bonus) Z_0 (mode impedance) at 4.0 GHz.

(c) Calculate the attenuation constant (in dB/cm) at 3.0 GHz. What is the attenuation (in dB) if the guide is 12 cm long? (Hint: Calculate the propagation constant first)

(d) (Bonus) What is the attenuation at frequencies much less than the TE_{10} cutoff frequency?

Hint: Look up what happens to the waves that can't propagate when the frequency of operation is less than the cut-off frequency. Do they get attenuated? Yes! But how?

2. A parallel-plates waveguide has a height of 2 cm and is filled with a dielectric of relative permittivity 2.25. The wave propagates in the z-direction as shown below. The magnetic field inside the waveguide is characterized by

$$H_z = 50\cos(100\pi x)\cos\left(3\pi \times 10^{10}t - \beta z\right)\ A/m$$ TE

Determine:

1. The mode that is propagating TE_2
2. The Phase constant

$TE \rightarrow E_z = 0$; $TM \rightarrow H_z = 0$

$H_z = \cos\left(\frac{m\pi x}{a}\right)$ $a = \frac{1}{50} \Rightarrow m = 2$

$\cos\theta = \frac{m\lambda}{2d} = \frac{2 \cdot 3\times10^8}{1.5 \times 3 \times 10^{10} \times 2 \times 1/100} = \frac{2}{3} \Rightarrow \sin\theta = \frac{\sqrt{5}}{3}$

$\beta = \frac{2\pi \times 1.5 \times 3\times10^{10} \times \sqrt{5}}{3\times10^8 \times 2 \times 3} = 50\pi\sqrt{5} = 351.2\ rad/m$

8. A parallel plate waveguide has a separation $d = 1$ cm, and is filled with teflon, having dielectric constant $\epsilon_r = 2.1$. Determine the maximum operating frequency such that only TEM mode will propagate. Also find the range of frequencies over which TE_1 and TM_1 (m=1) modes will propagate, and no higher order modes.

$d = 1/100$ $\epsilon_r = 2.1 \Rightarrow f_c = \frac{mv_p}{2d}$

TEM $m=0 \Rightarrow f_c = 0$ (propogates at any freq)

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